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Diabetic Retinopathy Detection using YOLO Model

Workshop Project by **Nilotpal Bose**
Practical AI

Introduction

- Diabetic Retinopathy (DR) is a diabetes complication that affects the eyes.
- Early detection is crucial to prevent vision loss.
- Traditional manual diagnosis is time-consuming and requires expertise.
- Deep learning models like YOLO automate this detection efficiently.

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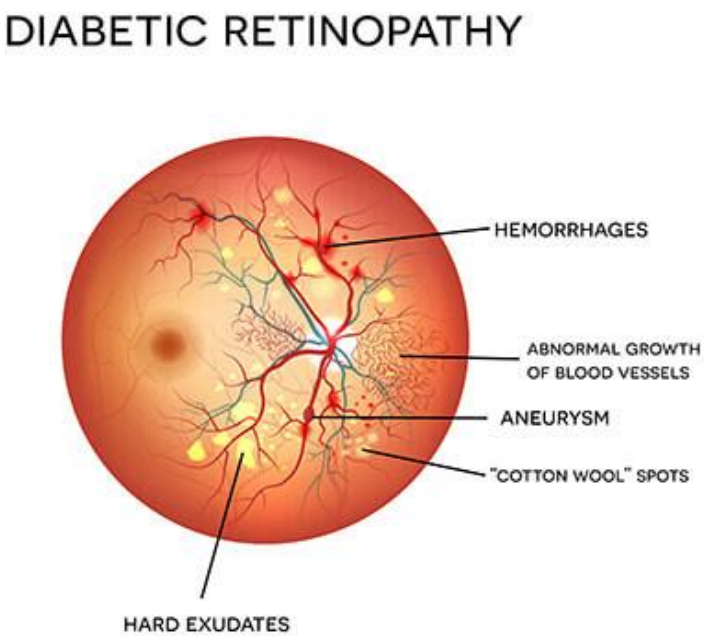
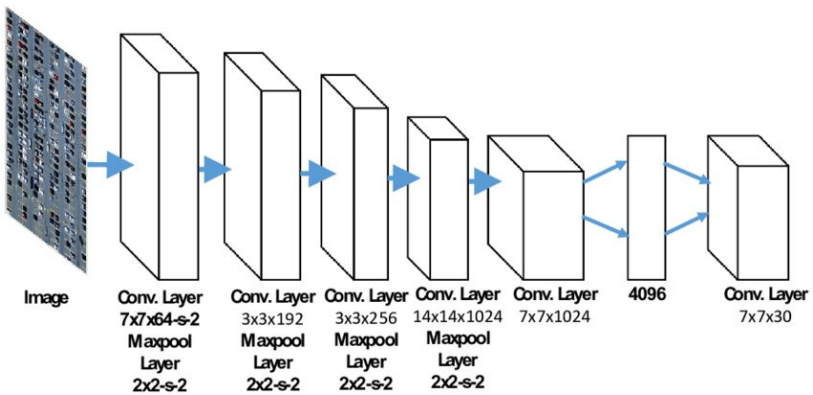
Dataset and Preprocessing

- Public datasets: Kaggle, EyePACS, APTOS, Messidor.
- Contains retinal fundus images labeled by Diabetic Retinopathy severity classes.
- Steps:
  - Image resizing and normalization
  - Data augmentation (rotation, zoom, brightness)
  - Label preparation for YOLO bounding boxes.



YOLO Model Overview



- YOLO (You Only Look Once) is a real-time object detection algorithm.
- Divides the image into grids and predicts bounding boxes and class probabilities.
- For Diabetic Retinopathy, it detects and classifies lesions (microaneurysms, hemorrhages, exudates, etc.).
- Advantages:

  - Fast inference
  - High accuracy

How YOLO Identifies Diabetic Retinopathy

Step 1: Input Retina Image

A fundus (retina) image is captured from the patient's eye.

Step 2: Image Division into Grids

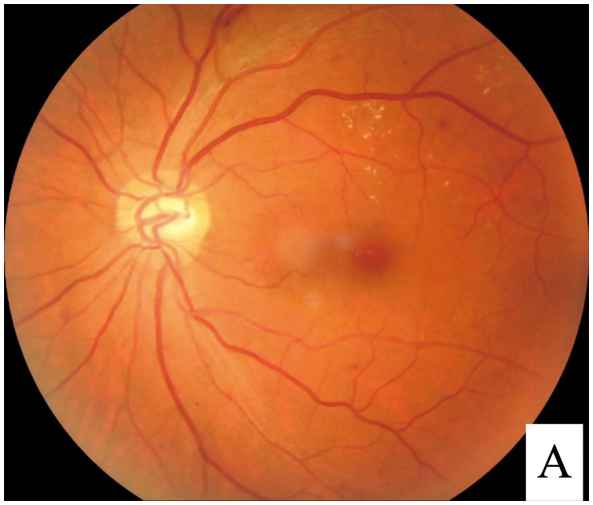
YOLO divides the image into grids. Each grid cell predicts:

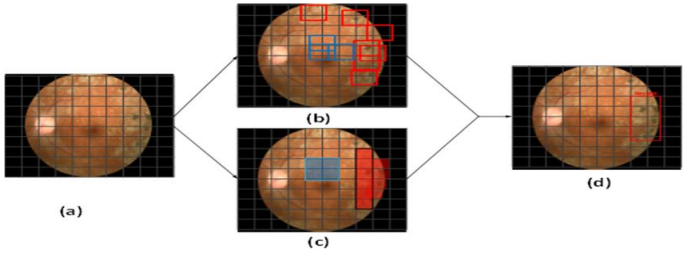
- Bounding box coordinates
- Confidence score
- Class probability (type of lesion)

Step 3: Lesion Detection

YOLO detects features associated with Diabetic Retinopathy, such as:

- Microaneurysms
- Hemorrhages
- Exudates (hard or soft)
- Cotton wool spots





How YOLO Identifies Diabetic Retinopathy

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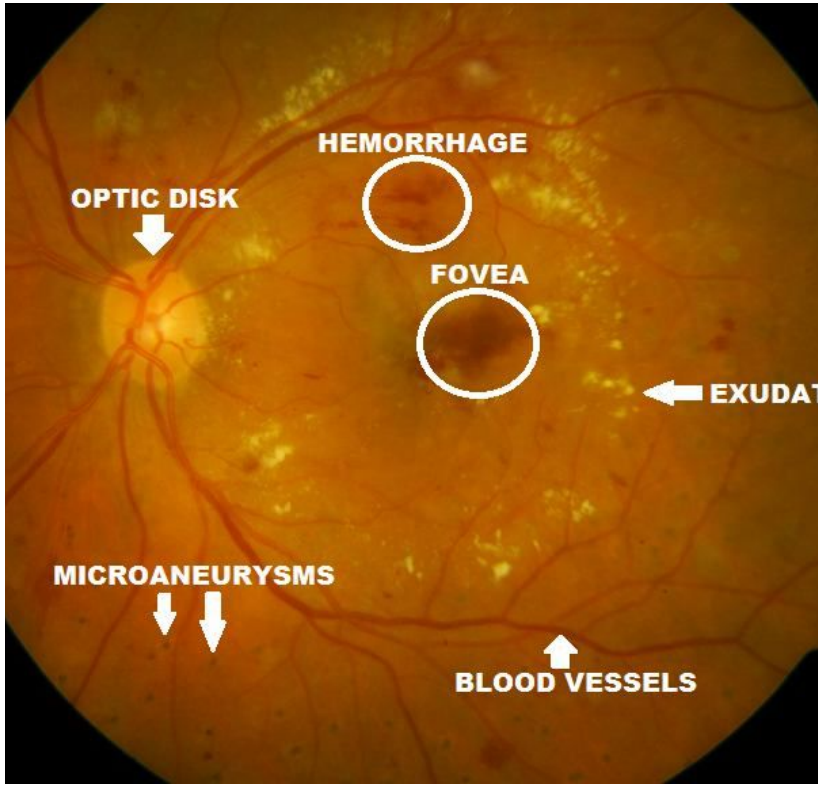
Step 4: Classification
Based on the number and severity of detected lesions, the image is classified into stages (e.g., No DR, Mild, Moderate, Severe, Proliferative).

Step 5: Output Visualization
Bounding boxes highlight lesion locations, and the model outputs a diagnostic category.



Hypothesis

- **Hypothesis:**
  YOLO identifies Diabetic Retinopathy by detecting small, localized abnormalities such as microaneurysms, hemorrhages, and exudates based on their distinct color, texture, and contrast patterns within retinal images.

- **Explanation:**
  Since YOLO divides the image into grids and predicts bounding boxes for objects in each region, it can learn to recognize these lesions as “objects” — bright (exudates - fluid that leaks out of blood vessels into nearby tissues) or dark (hemorrhages - high blood sugar damaging retinal blood vessels, causing them to swell, leak, or close) spots — with unique color and texture signatures that differ from normal retinal tissue.

Implementation and Results

- Framework: YOLO Object Detection Framework
- Training: ~50 epochs on GPU, learning rate tuning
- Metrics: Precision, Recall, mAP
- Observations:

  - YOLO achieved good accuracy for early Diabetic Retinopathy detection
  - Real-time performance suitable for clinical use.

Conclusion and Future Scope

- YOLO provides an efficient way to detect Diabetic Retinopathy automatically.
- Early diagnosis can significantly reduce blindness risk.
- Future Work:
  - Use larger datasets
  - Integrate with cloud healthcare systems
  - Explore hybrid models for better accuracy.